

IA Tripos Paper 4 – Mathematical Methods

2016 – CRIBS

Section A:

- (1) (a) Use l'Hopital's rule: twice on $f(x) = (x \cos x - \sin x)/x^3$

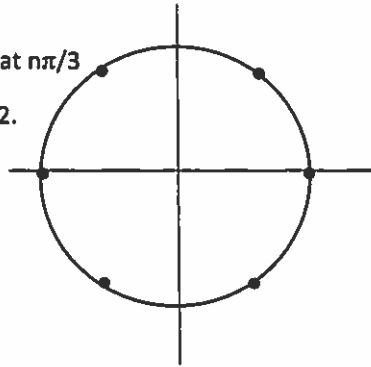
$$\text{First} = (\cos x - \sin x - \cos x)/3x^2 = -\sin x/3x^2$$

$$\text{Second} = [-\cos x/3] = -1/3$$

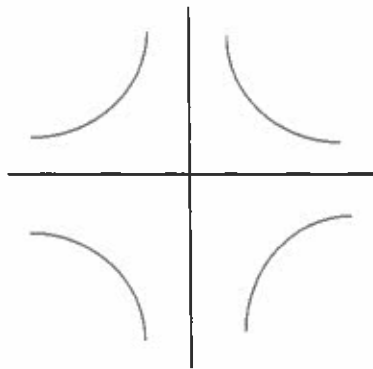
- (b) Use l'Hopital's rule once on $g(x) = \sin(\pi x)/(x^2 - 16)$

$$= \pi \cos(\pi x) / 2x = \pi \cos(-4\pi) / -8 = -\pi/8$$

- 2: (a) Solutions at $n\pi/3$
Circle radius 2.



- (b) $\text{Im}(z^2) = 2xy$



- (3) (a) In any pairs, the vectors **a**, **b**, **c**, satisfy $\mathbf{u} \cdot \mathbf{v} = 0$ so they are orthogonal.

Their magnitudes are $3\sqrt{2}$, 3, $3\sqrt{2}$ respectively so the volume is 56.

(b) by inspection the order they are in forms a right handed set as does any cyclic permutation.

(4) (a) By inspection U is a rotation of θ about the origin. For $\theta = \pi/4$ we get a 45° rotation.

$$[1,1] \rightarrow [\sqrt{2}, 0], [1,2] \rightarrow [3/\sqrt{2}, 1/\sqrt{2}], [2,2] \rightarrow [2\sqrt{2}, 0], [2,1] \rightarrow [3/\sqrt{2}, -1/\sqrt{2}]$$

Plot points and observe they form a square of unit side.

(b) $U^4 = 4$ rotations of -45° = one rotation of -1 , so $U^4 = -I$, can do it by numbers and multiply out the matrices. By inspection, rotation of $-\theta$ gives a transpose of U so $U^{-1} = U^T$

(c) $||A - \lambda I|| = 0$ implies $\lambda^3 - 3\lambda a^2 - 2a^3 = 0$ or $\lambda = -a, -a, 2a$.

largest eigenvalue: $\lambda = 2a$ Eigenvector: $(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$

Other two eigenvectors in the plane perpendicular to $\{1,1,1\}$.

(5) (a) $dN/dt = -\lambda N$ integrates up to $N(t) = N_0 \exp(-\lambda t)$

After time $T_{1/2}$ we have half the number of atoms left so $N(T_{1/2})/N_0 = \exp(-\lambda T_{1/2}) = 0.5$ and

$$\text{So : } T_{1/2} = \ln(2)/\lambda.$$

(b) If the N_1 species is stable, i.e. has a very long lifetime, then the N_1 's being produced are all those that decayed, $N_1(t) + N(t) = N_0$, so $N_1(t) = N_0 (1 - \exp(-\lambda t))$. This satisfies the boundary conditions that there are no N_1 species at $t=0$ and they are all N_1 species at very large time.

(c) If the N_1 species are unstable, then the rate of growth of N_1 is determined by the rate of formation of N_1 from N_0 minus the loss of N_1 to the daughter product N_2

$$\text{So: } dN_1(t)/dt = \lambda N_0 \exp(-\lambda t) - \lambda^* N_1(t) =$$

$$\text{Integrate } dN_1(t)/dt + \lambda^* N_1(t) = \lambda N_0 \exp(-\lambda t)$$

$$\text{Particular integral: } N_1(t) = A \exp(-\lambda t): -\lambda A + \lambda^* A = \lambda N_0, \text{ and so } A = \lambda N_0 / (\lambda - \lambda^*)$$

$$\text{No: } N_1(t) = A \exp(-\lambda t) + B \exp(-\lambda^* t), \text{ and given that } N_1(t=0) = 0, B = -A$$

$$N_1(t) = [N_0 / (\lambda - \lambda^*)] [\exp(-\lambda t) - \exp(-\lambda^* t)].$$

(d) $dN_1(t)/dt = 0$ when $\lambda \exp(-\lambda t) = \lambda^* \exp(-\lambda^* t)$ or $t^* = \ln(\lambda/\lambda^*) / (\lambda - \lambda^*)$

Note that both numerator and denominator are positive if $\lambda > \lambda^*$, and both are negative if $\lambda < \lambda^*$, so t^* is always positive.

PART A OF PAPER 4 OF THE PART IA TRIPOS 2016

Answers:

- (1) (a) $-1/3$
(b) $-\pi/8$
- (2) (a) $z=2\exp(in\pi/3)$ for $n=0,5$
(b) hyperbolae, $xy=\text{constant}$ (constant may be negative as well as positive)
- (3) Value =54, and a,b,c as given for a right-handed set.
- (4) (a) By inspection U is a -45 degree rotation in the plane about the origin.
(c) Eigenvalues $-a, -a, 2a$ Eigenvector for $+2a$ is $(1,1,1)/\sqrt{3}$ and the other two eigenvectors are any two perpendicular vectors in the plane perpendicular to $(1,1,1)$
- (5) (b) $N_1(t)=N_0(1 - \exp(-\lambda t))$
(c) $N_1(t) = [\lambda/(\lambda - \lambda^*)]N_0[\exp(-\lambda t) - \exp(-\lambda^*t)]$
(d) $t^* = \ln(\lambda/\lambda^*)/(\lambda-\lambda^*)$ positive whether $\lambda > \lambda^*$ or $\lambda < \lambda^*$

Q6 (Short)

Section - B

③

$$\frac{d^2x}{dt^2} - 4x = \sinh t$$

$$x(0) = 0$$

$$\left. \frac{dx}{dt} \right|_{x=0} = 0.$$

$$s^2 x - 4x = \frac{1}{s^2 - 1}$$

$$X(s) = \frac{1}{(s^2 - 1)(s^2 - 4)} = \frac{As + B}{(s^2 - 1)} + \frac{Cs + D}{(s^2 - 4)}$$

$$\Rightarrow As^3 + Bs^2 - 4As - 4B + Cs^3 + Ds^2 - Cs - D = 1$$

$$\Rightarrow A + C = 0 \quad -4A - C = 0 \quad \Rightarrow \boxed{A = C = 0}$$

$$B + D = 0 \quad -4B - D = 1$$

$$\Rightarrow \boxed{B = -\frac{1}{3} \text{ \& \; } D = \frac{1}{3}}$$

$$\therefore X(s) = \frac{-\frac{1}{3}}{(s^2 - 1)} + \frac{\frac{1}{3}}{(s^2 - 4)}$$

$$\Rightarrow \boxed{x(t) = \frac{1}{6} \sinh 2t - \frac{1}{3} \sinh t.}$$

Q7 (long)

④

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 5y = f(t); \quad y(0) = \dot{y}(0) = 0.$$

(a) Step response: $f(t) = 1 \quad t \geq 0.$

C.F.: $y = e^{\lambda t}, \quad f(t) = 0.$

$$\lambda^2 + 2\lambda + 5 = 0 \Rightarrow \lambda = \frac{-2 \pm \sqrt{4 - 20}}{2} \\ = -1 \pm 2i$$

$$\therefore y(t) = e^{-t} [A \cos 2t + B \sin 2t]$$

P.I.: $y = \alpha \Rightarrow 5\alpha = 1 \Rightarrow \alpha = 1/5$

$\therefore$ Step response is

$$y(t) = \frac{1}{5} + e^{-t} [A \cos 2t + B \sin 2t]$$

$$\dot{y}(t) = e^{-t} [(B - 2A) \cos 2t + (B + 2A) \sin 2t]$$

$$y(0) = 0 \Rightarrow \frac{1}{5} + A = 0 \Rightarrow \boxed{A = -1/5}$$

$$\dot{y}(0) = 0 \Rightarrow 2B - A = 0 \Rightarrow \boxed{B = -1/10}$$

$\therefore$ Step response is

$$y(t) = \frac{1}{5} - e^{-t} \left[\frac{\cos 2t}{5} + \frac{\sin 2t}{10} \right]$$

Impulse response is the derivative of the 5
Step response.

$$g(t) = \dot{y}(t) = \frac{\sin 2t}{2} e^{-t}.$$

(b) Now for $f(t) = \begin{cases} 0 & t < 0 \\ e^{-\alpha t} & t \geq 0. \end{cases}$

$$y(t) = \int_0^t g(t-\tau) f(\tau) d\tau$$

$$= \frac{1}{2} \int_0^t e^{-(t-\tau)} \sin 2(t-\tau) e^{-\alpha \tau} d\tau$$

$$= \frac{e^{-t}}{2} \int_0^t e^{(1-\alpha)\tau} \sin 2(t-\tau) d\tau$$

integrate by parts:

$$= \frac{e^{-t}}{2} \int_0^t \sin 2(t-\tau) d \left[\frac{e^{(1-\alpha)\tau}}{(1-\alpha)} \right]$$

$$= \frac{e^{-t}}{2} \left\{ \frac{e^{(1-\alpha)\tau}}{(1-\alpha)} \sin 2(t-\tau) \right\}_0^t + \frac{2}{(1-\alpha)} \int_0^t e^{(1-\alpha)\tau} \cos 2(t-\tau) d\tau$$

$$= \frac{e^{-t}}{2} \left\{ \frac{-\sin 2t}{(1-\alpha)} + \frac{2}{(1-\alpha)} \int_0^t \cos 2(t-\tau) e^{(1-\alpha)\tau} d\tau \right\} \quad (6)$$

$$= \frac{e^{-t}}{2} \left\{ \frac{-\sin 2t}{(1-\alpha)} + \frac{2}{(1-\alpha)} \int_0^t \cos 2(t-\tau) d \left[\frac{e^{(1-\alpha)\tau}}{(1-\alpha)} \right] \right\}$$

$$= \frac{-e^{-t} \sin 2t}{2(1-\alpha)} + \frac{e^{-t}}{2} \left\{ \frac{2}{(1-\alpha)^2} \left[e^{(1-\alpha)\tau} \cos 2(t-\tau) \right]_0^t - 2 \int_0^t e^{(1-\alpha)\tau} \sin 2(t-\tau) d\tau \right\}$$

$$= -\frac{e^{-t} \sin 2t}{2(1-\alpha)} + \frac{e^{-t}}{(1-\alpha)^2} \left[e^{(1-\alpha)t} - \cos 2t \right] - \frac{4}{(1-\alpha)^2} \underbrace{\frac{e^{-t}}{2} \int_0^t e^{(1-\alpha)\tau} \sin 2(t-\tau) d\tau}_{y(t)}$$

$$\Rightarrow \left(1 + \frac{4}{(1-\alpha)^2} \right) y(t) = \frac{-e^{-t} \sin 2t}{2(1-\alpha)} + \frac{e^{-\alpha t} - e^{-t} \cos 2t}{(1-\alpha)^2}$$

$$\therefore y(t) = \left[1 + \frac{4}{(1-\alpha)^2} \right]^{-1} \left[\frac{e^{-\alpha t} - e^{-t} \cos 2t}{(1-\alpha)^2} - \frac{e^{-t} \sin 2t}{2(1-\alpha)} \right]$$

or

$$y(t) = \left[1 + \frac{(1-\alpha)^2}{4} \right]^{-1} \left[\frac{e^{-\alpha t} - e^{-t} \cos 2t}{4} - \frac{(1-\alpha) e^{-t} \sin 2t}{8} \right]$$

If the integration by parts is done using $\sin 2(t-\tau)$

(7)

When $\alpha \rightarrow 0$

both of the above solutions give

$$y(t) = \frac{1}{5} - e^{-t} \left[\frac{\cos 2t}{5} + \frac{\sin 2t}{10} \right]$$

Q8 (short)
(a)

$$P_0 = 1 - (P_1 + P_2 + P_3 + \dots)$$

$$= 1 - (\alpha p + \alpha p^2 + \alpha p^3 + \dots)$$

$$\boxed{P_0 = 1 - \frac{\alpha p}{1-p}}$$

(Geometric Series
- from data book)

as required.

$$(b) \quad (1-x)^{-(n+1)} = 1 - (n+1)(-x) + \frac{-(n+1)(-n-2)}{2!}(-x)^2 + \dots$$

(Binomial expansion
from the data book)

$$= 1 + (n+1)x + \frac{(n+1)(n+2)}{2!}x^2 + \dots$$

$$\frac{1}{(1-x)^{n+1}} = \binom{n}{n} + \binom{n+1}{n}x + \binom{n+2}{n}x^2 + \dots$$

as required.

Q9 (Short)

$$f(x, y) = 5x^3 - 2y^3 - 6x^2y + 54y$$

$$\frac{\partial f}{\partial x} = 15x^2 - 12xy = 3x(5x - 4y) = 0$$

$$\frac{\partial^2 f}{\partial x^2} = 30x - 12y$$

$$\Rightarrow \begin{cases} x = 0 \\ x = \frac{4}{5}y \end{cases}$$

$$\frac{\partial f}{\partial y} = -6y^2 - 6x^2 + 54 = -6(x^2 + y^2) + 54 = 0$$

$$\Rightarrow x^2 + y^2 = \frac{54}{6} = 3^2 \quad \text{--- (1)}$$

$$\frac{\partial^2 f}{\partial y^2} = -12y$$

$$x = 0 \text{ in (1)} \Rightarrow y = \pm 3$$

$$x = \frac{4}{5}y \text{ in (1)} \Rightarrow y = \pm \frac{15}{\sqrt{41}}$$

$$\frac{\partial^2 f}{\partial x \partial y} = -12x$$

Stationary points are

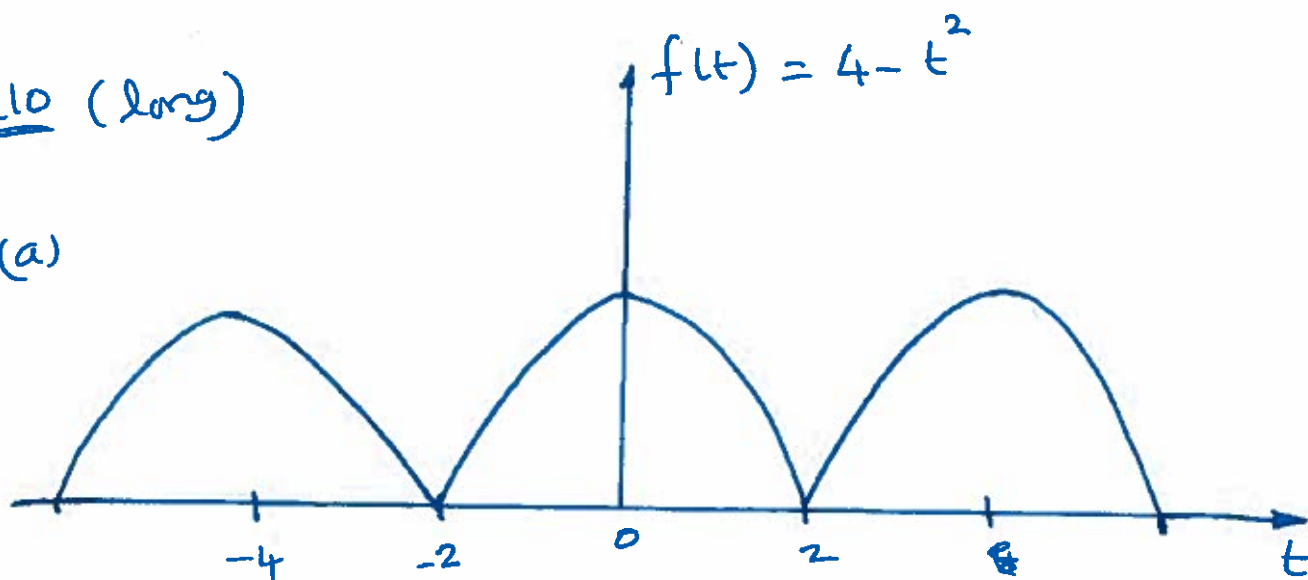
$$(0, 3), (0, -3), \left(\frac{12}{\sqrt{41}}, \frac{15}{\sqrt{41}}\right), \left(-\frac{12}{\sqrt{41}}, -\frac{15}{\sqrt{41}}\right)$$

$$\Delta = \frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \left[\frac{\partial^2 f}{\partial x \partial y} \right]^2$$

	$\frac{\partial^2 f}{\partial x^2}$	$\frac{\partial^2 f}{\partial y^2}$	$\frac{\partial^2 f}{\partial x \partial y}$	Δ	
$(0, 3)$	-36	-36	0	+ve	<u>Maximum</u>
$(0, -3)$	36	36	0	+ve	<u>Minimum</u>
$\left(\frac{12}{\sqrt{41}}, \frac{15}{\sqrt{41}}\right)$	$\frac{180}{\sqrt{41}}$	$-\frac{180}{\sqrt{41}}$	$-\frac{144}{\sqrt{41}}$	-ve	<u>Saddle</u>
$\left(-\frac{12}{\sqrt{41}}, -\frac{15}{\sqrt{41}}\right)$	$-\frac{180}{\sqrt{41}}$	$\frac{180}{\sqrt{41}}$	$\frac{144}{\sqrt{41}}$	-ve	<u>Saddle</u>

Q10 (long)

(a)



$$T = 4, \quad (-2, \text{to } 2)$$

$$(b) \quad f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi n t}{T} \quad (\text{even function})$$

$$a_n = \frac{2}{T} \int_{-2}^2 f(t) \cos\left(\frac{2\pi n t}{T}\right) dt$$

$$a_0 = \frac{1}{2} \int_{-2}^2 f(t) dt = \int_0^2 f(t) dt$$

$$= \left[4t - \frac{t^3}{3} \right]_0^2 = \frac{16}{3}$$

$$\boxed{a_0 = \frac{16}{3}}$$

(10)

$$a_n = \frac{1}{2} \int_{-2}^2 (4-t^2) \cos\left(\frac{n\pi t}{2}\right) dt$$

integrate by parts:

$$= \frac{1}{2} \left[\frac{8}{n\pi} \sin\left(\frac{n\pi t}{2}\right) \right]_{-2}^2 - \frac{1}{2} \int_{-2}^2 t^2 \cos\left(\frac{n\pi t}{2}\right) dt$$

$$= -\frac{1}{2} \int_{-2}^2 t^2 d \left[\frac{\sin(n\pi t/2)}{n\pi/2} \right]$$

$$= -\frac{1}{n\pi} \left[t^2 \sin\left(\frac{n\pi t}{2}\right) \right]_{-2}^2 + \frac{1}{n\pi} \int_{-2}^2 \sin\left(\frac{n\pi t}{2}\right) (2t) dt$$

$$= \frac{2}{n\pi} \int_{-2}^2 t d \left[\frac{-\cos(n\pi t/2)}{n\pi/2} \right]$$

$$= -\frac{4}{n^2\pi^2} \left[t \cos\left(\frac{n\pi t}{2}\right) \right]_{-2}^2 + \frac{4}{n^2\pi^2} \int_{-2}^2 \cos \frac{n\pi t}{2} dt$$

$$= -\frac{4}{n^2\pi^2} [2 \cos n\pi + 2 \cos n\pi] + \frac{4}{n^2\pi^2} \frac{2}{n\pi} \left[\sin\left(\frac{n\pi t}{2}\right) \right]_{-2}^2$$

$$a_n = -\frac{16}{n^2\pi^2} (-1)^n = \frac{16}{n^2\pi^2} (-1)^{n+1} \quad n \neq 0.$$

(11)

$$\therefore f(t) = \frac{8}{3} + \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} \cos\left(\frac{n\pi t}{2}\right)$$

as required.

$$(c) \quad f(0) = 4 = \frac{8}{3} + \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$$

$$1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$$

as required.

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Solutions to Section C

11. Data structures

(a) Any one of the following four lines is acceptable:

```
board[0][0].type = Rook; board[0][0].colour = White;
board[0][7].type = Rook; board[0][7].colour = White;
board[7][0].type = Rook; board[7][0].colour = White;
board[7][7].type = Rook; board[7][7].colour = White;
```

[2]

(b)

```
for (int i=0; i<8; i++)
  for (int j=0; j<8; j++)
    if (board[i][j].type == Pawn) cout << i << ' ' << j << endl;
```

[4]

(c)

```
int n = 0;
for (int i=0; i<8; i++)
  for (int j=0; j<8; j++)
    if ((board[i][j].type != None) and (board[i][j].colour == Black)) n = n+1;
cout << n << endl;
```

[4]

12. Algorithmic complexity and numerical accuracy

(a) f_1 is clearly $O(n)$. The complexity of f_2 depends on the implementation of $\exp$, though we can expect this to be fast on common architectures. In fact, the standard x86 implementation of $\log$ is $O(1)$ and appears to use more or less this method (there are single x86 machine code instructions for exponentiation and scaled logarithms). [4]

(b) The program will display a small, non-zero number. This is because f_1 will calculate 7^2 precisely, whereas f_2 will lose some precision in the floating point representation of $\log(7.0)$. The main program does, in fact, display $7.62939\text{e-}06$. [3]

(c) f_1 will fail when n is negative, f_2 will fail when a is zero or negative. Both will fail when a and/or n are sufficiently large to cause a floating point overflow. [3]